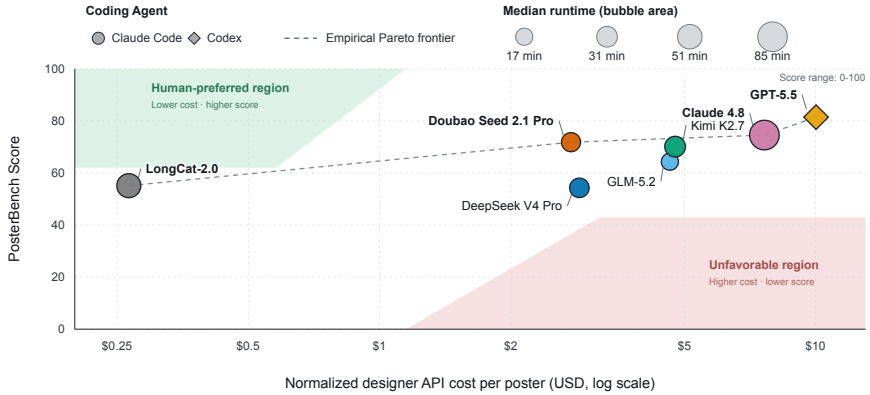


Figure 8 Cost–performance trade-off across seven AutoDesign model configurations on PosterBench-mini, the fixed 10-paper subset; marker size denotes median runtime, and the dashed line denotes the empirical Pareto frontier.

67.6% against Claude Design.

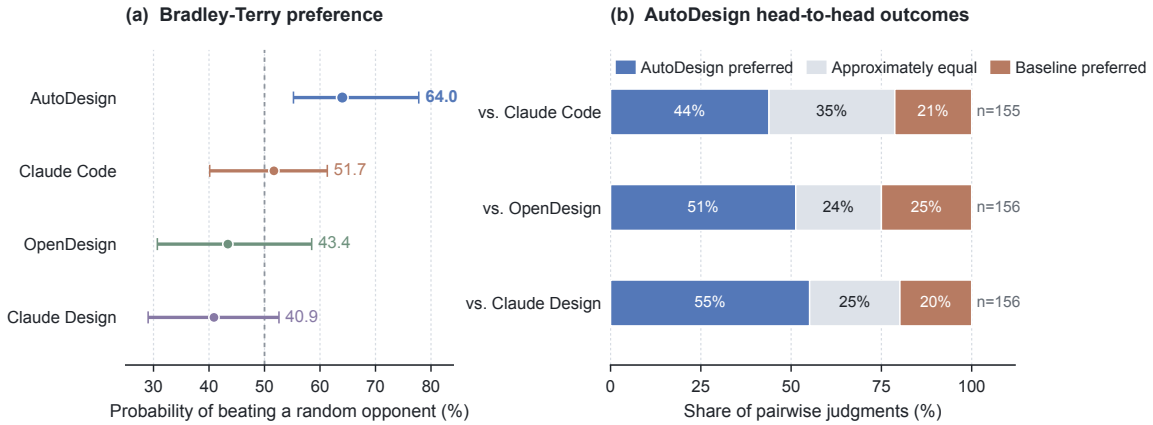


Figure 9 System-blind human evaluation. (a) Bradley–Terry probability of beating a uniformly sampled alternative, with 95% intervals crossed by paper–reviewer bootstrap replicates. (b) AutoDesign head-to-head outcomes against each baseline. Eleven volunteer reviewers submitted 936 responses, including 933 ranking judgments and three skips.

Benchmark–human alignment. The blind study also assesses whether PosterBench provides an informative comparison signal at the poster level. Figure 10(a) compares each poster's PosterBench Score with its tie-adjusted human preference. Each human score aggregates the blind pairwise judgments involving that poster. The two measurements have a positive, albeit imperfect, association ($r = 0.34$; a paper-cluster bootstrap gives a 95% interval of $[0.22, 0.44]$). This is a useful property of the protocol rather than a requirement that it duplicate human preference: PosterBench also evaluates Faithfulness, Coverage, Density, Visual Evidence, Layout, Readability, and Aesthetics, whereas each blind judgment asks for an immediate pairwise choice.

More importantly, the PosterBench Score margin calibrates how informative an automatic comparison is. Figure 10(b) uses 919 of the 933 non-skip judgments: the remaining 14 assign equal PosterBench Scores to both posters and therefore have no benchmark-preferred direction. The probability that the benchmark-preferred poster agrees with the human decision rises from 51.9% for 0–3-point gaps to 74.4% for gaps of at least 20 points. The error bars are 95% intervals from 2,000 crossed paper–reviewer bootstrap replicates, which resample both source papers and reviewers. Thus, the benchmark offers more than a global system ranking: a large score gap identifies comparisons in which human preference is substantially more consistent.

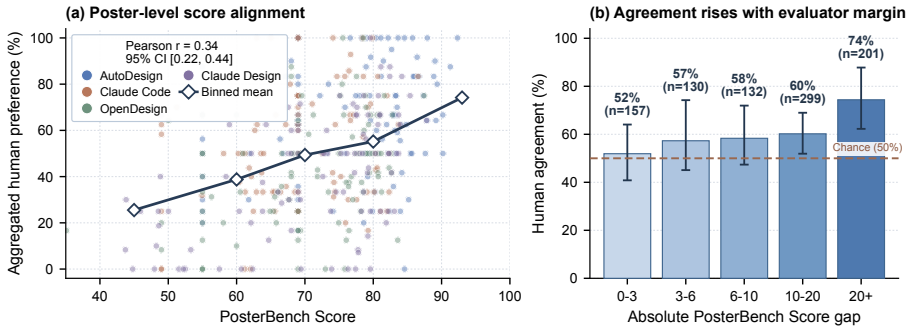


Figure 10 PosterBench Score alignment with system-blind human preference. (a) Poster-level score association. (b) Human agreement increases with the PosterBench Score margin.

5.4 Qualitative Analysis of Designer's Trajectory

Figure 11 traces one poster run across five attempts. The critic first identifies a clipped analysis lane at A1 (0.36); the reallocation of the row removes the constraint at A3 (0.42), and later the refit of the header and the scaling of evidence produce a more balanced hierarchy at A5/A6 (0.62). A9 preserves the repaired composition and is accepted at 0.78. The trace illustrates the intended use of diagnostic feedback: edits stay localized to the failing region while valid layout and source-derived content are retained across revisions.

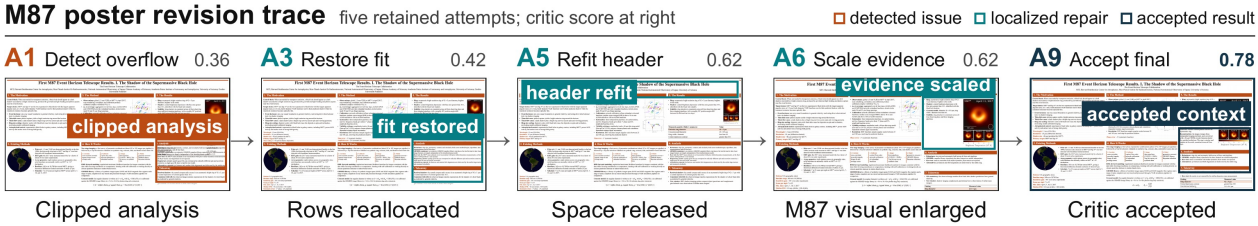


Figure 11 Qualitative trajectory of a poster generation run. Five selected attempts show the local change associated with each diagnostic or repair event. Colored boxes mark the implicated region: an analysis overflow at A1, its fit restoration at A3, header refitting at A5, evidence scaling at A6, and the critic-accepted final at A9.

6 Future Directions

AutoDesign is presently validated for academic paper-to-poster generation, but the underlying agentic design pattern is not tied to a single input or output medium. The pilot artifacts in Figure 13 show that the current DesignHarness can also produce paper-to-slide, paper-to-webpage, and paper-to-conference-video outputs. Figure 12 illustrates the long-term direction: a multimodal-in, multimodal-out agentic design system that integrates papers, visual evidence, code, data, and human guidance, then iteratively creates the medium-appropriate output for a target communication setting. Making this expansion reliable requires more than exposing new output formats. Each medium needs source–output data, an evaluator, a rendering and validation gate, and an objective tailored to its communication setting. PosterBench formally evaluates academic posters only; the slide, webpage, and video artifacts therefore remain pilots. Shared context construction, preference memories, and repair histories could nevertheless provide a substrate for reusing experience across media, provided that the transfer is evaluated against medium-specific objectives. At meta-harness level, better component selection and evaluator evolution remain open problems. A selector should choose next bounded update from failure attribution, uncertainty, expected improvement, and component interactions. Any adaptive evaluator must remain versioned and anchored by frozen reference tasks, adversarial probes, and periodic human audits so optimization doesn't reward-hack a moving target.

on Machine Learning, volume 267 of *Proceedings of Machine Learning Research*, pages 80569–80611, 2025. https://proceedings.mlr.press/v267/zhuge25a.html.